

Integrating Mechanisms for Insulin Resistance: Common Threads and Missing Links

Insulin resistance is a complex metabolic disorder that defies a single etiological pathway. Accumulation of ectopic lipid metabolites, activation of the unfolded protein response (UPR) pathway and innate immune pathways have all been implicated in the pathogenesis of insulin resistance. However, these pathways are also closely linked to changes in fatty acid uptake, lipogenesis, and energy expenditure that can impact ectopic lipid deposition. Ultimately, accumulation of specific lipid metabolites (diacylglycerols and/or ceramides) in liver and skeletal muscle, may be a common pathway leading to impaired insulin signaling and insulin resistance.